
GPT Caddy

A utility to make GPT Golf easier.
See inside the run.
Decide what to change next.

I built it for myself — every GPT Golf run goes through it.

FORMAT

SLIDES

LIVE RUN

Q&A

EVERY RUN, END TO END



Naive GPT Golf is guess-and-check.

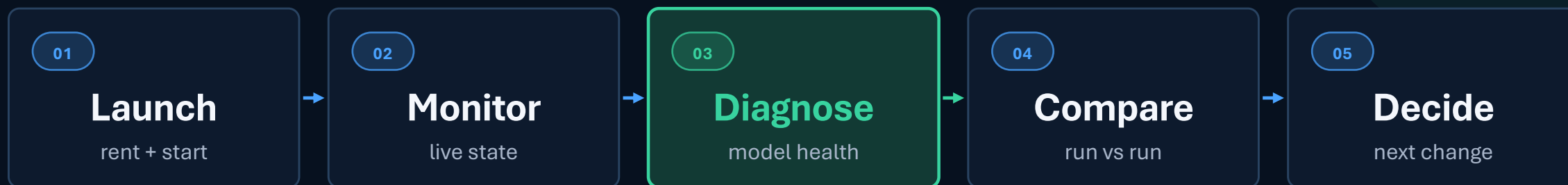
Tweak → wait → inspect loss → guess again?

Logs tell you what happened.

```
step:1000/20000 train_loss:3.0888 train_time:199487ms step_avg:199.48ms  
step:1250/20000 train_loss:3.0824 train_time:249358ms step_avg:199.49ms  
step:1500/20000 train_loss:3.0743 train_time:299232ms step_avg:199.49ms  
step:1750/20000 train_loss:3.0707 train_time:349106ms step_avg:199.50ms  
step:2000/20000 train_loss:3.0401 train_time:398978ms step_avg:199.49ms  
step:2250/20000 train_loss:3.0157 train_time:448839ms step_avg:199.45ms  
step:2500/20000 train_loss:2.9648 train_time:498704ms step_avg:199.46ms  
step:2750/20000 train_loss:2.8870 train_time:548568ms step_avg:199.46ms
```

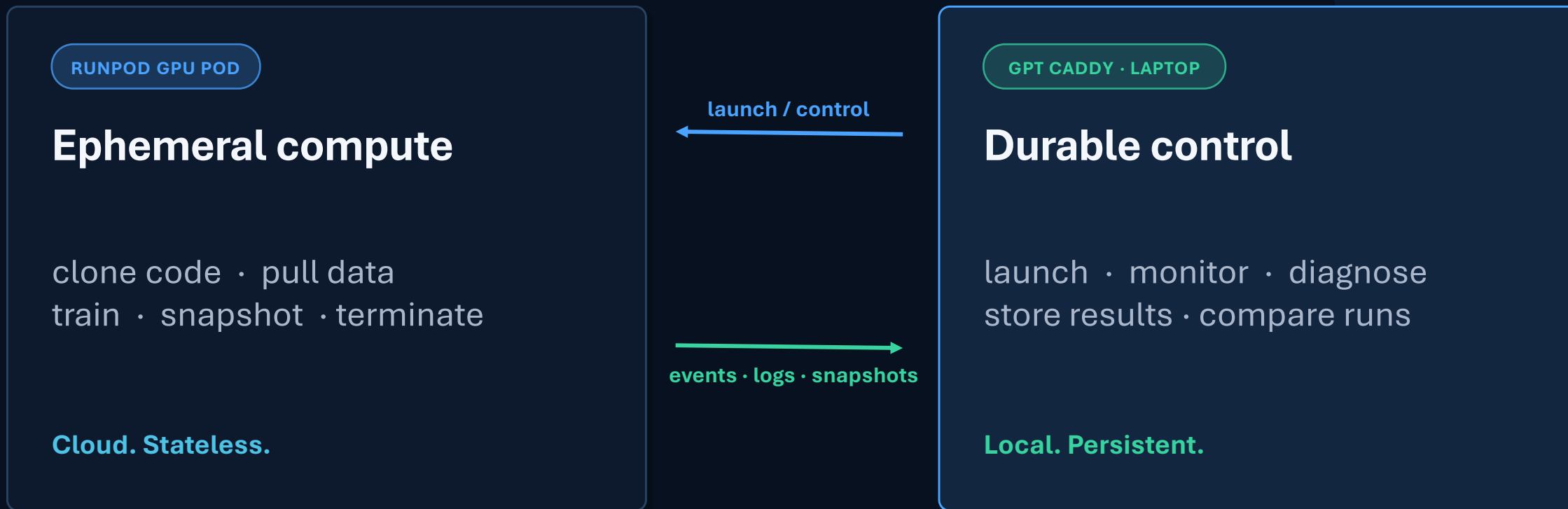
— not why.

Shrink the loop. Iterate faster.



Every pass through the loop should reduce uncertainty.

Ephemeral GPU. Durable control plane.



The pod can disappear. The durable run history survives.

Time to first token matters.

SENTENCEPIECE FORK

6x+ faster tokenization

>50GB Raw Text -> ~10B tokens -> 200MB Shards

Over 1 hour → 8-10 minutes

~50% less memory ~600% throughput

Parallel buffering + divide-and-conquer the corpus.

CUDA / LOADER

<2GB Container Faster starts.

Cut the stuff that isn't used

20% the size of "full" distro

16 KB stubs satisfy the loader

Derived from the installed Torch binary.
Restore missing deps for special cases.

Multi-tier build. Bake CUDA/Torch once. Add new deps in seconds.

A small event path keeps the live view honest.



- **Boots before dependencies**

stdlib-only daemon

- **Every event is durable**

persisted to SQLite

- **One stream, three uses**

browser · history · diagnostics

Pod → authenticate → laptop → browser / persistence / diagnostics / monitoring

LIVE

Watch three things move.

01

Race vs baseline

Ahead or behind at the same budget?

02

Cost in real time

Meter ticks; the pod stops itself.

03

Model health

Metrics live; probe runs off-GPU.



Keeps you informed without costing your focus.

DIAGNOSE

The diagnostics check their own arithmetic.

01

RE-COMPUTE

layer outputs

02

RE-DERIVE

loss independently

03

CROSS-CHECK

BPB vs on-pod eval

gg-main-0622-205629 · main step 1000 just now

WARN

Deep Dive

WARN

3 checks need review in capacity and flow.

4 healthy 3 warnings 0 failing 0 unavailable 2 info

Capacity

1 good · 2 warn

Dead neurons 7.2% Head redundancy 0.636

Attn entropy (min) 1.528

Flow

1 good · 1 warn · 1 info

Mean loss 2.778 Residual growth 0.375

Logit saturation 0.0%

Gradients

1 good

Grad norm ratio 11.436

Architecture

1 good · 1 info

Embed eff_rank 212.519 Zero-init escape (min) 4.281

1000

Latest

NEEDS ATTENTION

Dead neurons

Units that never activate waste capacity; a large fraction means dead width. · warn ≥ 0.05 , bad ≥ 0.2

7.2%

Head redundancy

Attention heads computing near-identical outputs are redundant capacity. · warn ≥ 0.6 , bad ≥ 0.85

0.636

Residual growth

Residual stream norm should grow moderately; too flat or exploding is unhealthy. · ok 0.5–4

0.375

GPU trains. Laptop audits. Monitor progress without slowing down.

COMPARE

Did my change help?



01

Did loss improve?

curves + no-regression banner

02

Did validation improve?

BPB at equal progress

03

Did model health improve?

verdict + headline metrics

Compare at equal step or equal progress. Then decide what to change next.

Launch → Monitor → Diagnose → **Compare** → Decide 🔄

CLOSE

From guess-and-check to informed decisions.

GPT Golf is an optimization problem without a *profiler*.
Then I **built** one.

need → tool → evidence → decision

Compute is temporary but the memories last forever.